

SEBASTIAN RENDON LARREA

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Probability- and expected-value-driven quantitative researcher who designs, backtests, and independently validates options and trading models in Python, with a focus on out-of-sample rigor and model risk.

EDUCATION

The University of Texas at El Paso, Woody L. Hunt College of Business

El Paso, TX

B.B.A., Financial Analysis & Information Systems | GPA 3.90/4.00 | Expected December 2026

- Dean's List every semester (Spring 2023-present); eBay Scholarship, Battle of the Brains competition winner.
- Completed the first year of the same degree at Universidad de Guadalajara, Mexico (2021-2022) before transferring to UTEP; semester abroad at Universidad del Norte, Colombia (Fall 2024).
- Quantitative coursework: Analysis of Derivatives, Investments, Portfolio Analysis, AI & ML Applications in Finance, Business Statistics & Analytics, Money & Banking.

QUANTITATIVE RESEARCH & TRADING MODELS

Systematic Options & Volatility Research (Python)

- Built systematic options strategies scored on expected value from first-passage (barrier-hit) probabilities under a lognormal diffusion (priced with Black-Scholes and binomial trees), and engineered an implied-volatility curvature ("kink") signal with cross-sectional and historical z-scoring to rank structures.
- Extended to event-driven earnings-volatility research (pre-earnings ATM-forward straddles; post-earnings defined-risk iron condors) with PEAD controls and point-in-time (anti-look-ahead) earnings calendars; evaluated via out-of-sample backtesting, transaction-cost modeling, Wilson-interval evidence floors, and VIX-regime stratification.
- Enforced model risk with a multi-agent research framework (independent implementer + challenger under my direction), pre-registered falsification gates, and 349 automated tests; stress-tested the closed-form first-passage model against Monte Carlo over a reproducible CBOE-SFTP data pipeline, surfacing a conservative, rank-preserving bias (Spearman $\rho \approx 0.99$).
- Open-sourced (CI-tested, arbitrage-free SSVI synthetic data) at github.com/sebasrenla/defined-risk-options-lab.

Systematic Equity Portfolio: Out-of-Sample Study

- Built a maximum-Sharpe (mean-variance) 10-stock portfolio (CAPM expected returns, portfolio beta/variance, Herfindahl concentration) on monthly WRDS/CRSP data (2019-2023).
- An out-of-sample test on 2024 exposed large in-sample-to-out-of-sample degradation vs. the S&P 500, a case study in overfitting, feature decay, and market-regime (Magnificent-Seven) concentration.

Derivatives Pricing & Statistical Modeling

- Built numerical option-pricing models (binomial trees with American exercise) and multi-stage equity valuations (CAPM with Hamada lever/unlever, WACC, FCFE/FCFF).
- Modeled financial time series with OLS regression (R^2 /F-tests) and built Monte Carlo simulations (dividend-growth and asset allocation) to distribution-test valuations and portfolio outcomes.

EXPERIENCE

DocuSign Developer, UTEP PeopleSoft

El Paso, TX · Oct 2023-Present

- Developed SQL queries and data extracts to analyze document-workflow activity and support reporting on approval volumes and processing trends.
- Automated financial document-approval workflows across 20+ departments, improving processing efficiency ~150%.

Budget & Payroll Analyst I, UTEP Budget & Payroll Services

El Paso, TX · Aug 2022-Oct 2023

- Built an Excel/VBA tool that replaced a manual process, reconciling ~300 monthly cost-center files by standardizing formats, matching salaries against prior months, and auto-flagging exceptions into per-owner review lists.
- Performed 150+ monthly budget analyses and advised 65+ departments on cost allocation and compliance.

Founder & Manager, Entregas y Más

Ciudad Juárez, MX · Jun 2020-Aug 2022

- Founded a pandemic-era delivery business at age 16 and grew monthly profit ~50% within three months through market research, a first exercise in decision-making under uncertainty.

TECHNICAL SKILLS

- **Quantitative:** probability & statistics, expected-value / probabilistic reasoning, stochastic modeling (lognormal/GBM, first-passage & barrier probabilities), numerical methods (binomial trees, Monte Carlo), backtesting (out-of-sample, transaction costs, feature decay), time-series & regression, portfolio optimization, options pricing & Greeks.
- **Programming & data:** Python (intermediate, pandas/NumPy), SQL, VBA; Git; reproducible research (checksummed data, automated testing); WRDS/CRSP/Compustat, Bloomberg, Capital IQ.
- **Programs & languages:** J.P. Morgan Quantitative Research virtual experience (Forage) · ALPFA; Banking Academy · Fully bilingual in English and Spanish.